

MATH 2790 — Exam 2 Review Worksheet

- (1) An annuity pays continuously for 12 years at a rate of payment $R(t) = 4t + 4$ and the force of interest is $\delta(t) = \frac{2}{t+1}$. Find the present value of this annuity.
- (2) Taylor purchases a small business for \$180,000 financing the balance over 8 years with *monthly payments* at an annual interest rate of 8.1%.
- (a) Find the level monthly payment. **Round up** your answer to 2 decimals.
- (b) Compute the outstanding loan balance immediately after 50 payments. Round to 2 decimals. Notice the final payment will be adjusted so try to use the retrospective method.
- (3) You deposit end of month payments of X into a bank account for 8 years so that starting in the 9th year you can withdraw Y every year for 15 years. Letting i be a quarterly effective interest rate, set up an equation relating X and Y using actuarial notation.
- (4) Ava purchases an increasing annuity-due that pays \$5 at the beginning of the first month, \$8 at the beginning of the second month, and continues to increase by \$3 per month thereafter. The last payment is \$86. The nominal annual rate is 9% convertible monthly. Find the present value of this annuity one month before the first payment. Round to two decimals.
- (5) You plans to deposit funds each year for 22 years to launch a start-up company right after the 22nd deposit.. The first deposit is \$2,500 and due to the excitement of the business, you increase each payment by 5% annually. Assume an annual effective rate of 6.2%. You want to throw a party exactly one year before the company launches. Find the value of *all* the payments during the time of the party.
- (6) Suppose \$40,000 was invested on January 1, 1980 at an annual effective interest rate of 7% in order to provide an annual (calendar-year) scholarship of \$5,000 each year forever, the scholarships paid out each January 1. In what year can the first \$5,000 scholarship be made?
- (7) Determine the present value of 1 payable at the end of years **7, 12, 17, 22, 27, 32**. Pick the correct choice and show your work to support your choice. Leave your answer in terms of only actuarial symbols $a_{\overline{n}|i}$, $s_{\overline{n}|i}$, $\ddot{a}_{\overline{n}|i}$, $\ddot{s}_{\overline{n}|i}$ where i is the annual effective interest rate.